

Asrith Pericherla

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SUMMARY: AI/ML Engineer experience in building and deploying **Gen AI/RAG solutions** using **LangChain**, **FAISS**, and **AWS Bedrock**. Expert in **unstructured data pipelines (Glue, Lambda)**, **prompt engineering**, and **agentic workflows (LangChain Agent API, AutoGen)** with **multi-cloud LLM integrations**. Proven track record of accelerating insights delivery through scalable, serverless architectures and Agile collaboration.

TECHNICAL SKILLS:

Languages & Libraries:	Python (NumPy, Pandas), Java(Spring Boot), SQL, Bash
ML / DL:	PyTorch, TensorFlow, sci-kit-learn, XGBoost, Prophet, Hugging Face Transformers, RLHF, Detectron2
Gen-AI & RAG:	LangChain, Vector DBs, OpenAI GPT-4o, Claude
NLP:	spaCy, NLTK, Rasa, BERT-family models
Data & Streaming:	Apache Kafka, Apache Spark (Structured Streaming), Apache Airflow, Redis, PostgreSQL, MongoDB
Cloud & MLOps:	AWS (SageMaker, Bedrock, EKS, Lambda, S3, Glue, AppConfig, CloudWatch, Model Monitor), Azure OpenAI, GCP Vertex AI, Docker, Kubernetes, Terraform, GitHub Actions
Experiment Tracking & Observability:	MLflow, Kubeflow, Prometheus, Grafana, ELK/Elastic
APIs & Services:	FastAPI, Flask, gRPC, REST, OpenAPI/Swagger, OAuth 2.0
Testing & Optimization:	Optuna, pytest, SonarQube, Bayesian hyperparameter search
Visualization & Reporting:	Grafana, QuickSight, Jupyter Notebooks, Confluence
Methodologies:	Agile (Scrum), Test-Driven Development (TDD), CI/CD Pipelines, MLOps

PROFESSIONAL EXPERIENCE:

- **Adient, US (Pennsylvania) – Software Engineer - AI/ML** **May 2023 – Present**
- Leveraged **AWS Bedrock Titan** and **Anthropic Claude** foundation models to generate narrative executive summaries for each application's **R lane**, **complexity**, and **cloud readiness** findings, automating report drafting and reducing manual write-ups by **60%**
 - Created a **RAG pipeline** by ingesting harmonized **ServiceNow**, **SCCM**, and **Zscaler** data into an **S3-backed knowledge base**, embedding document vectors with **FAISS**, and serving context to **Claude LLM** for accurate **Q&A** on application dependencies
 - Automated parsing of **CMDB** exports (**JSON**), **SCCM CSVs**, and **Zscaler logs** via **AWS Glue** and **Lambda (PySpark)**, normalizing and tokenizing text and metadata to ensure clean inputs for summarization and analysis by **Titan** or **Claude**
 - Designed, tested, and iterated **prompts** for **Bedrock-hosted Titan** and **Claude** models, tuning **temperature**, **max tokens**, and example **shots** to reduce **hallucinations by 25%** when generating **R lane justifications** and **dependency narratives**

- Built **agentic AI workflows** using **LangChain's Agent API** and **AutoGen** hosted on **AWS Lambda** to orchestrate multi-step tasks—ingesting data via **Athena**, performing **FAISS vector retrieval**, and invoking **Bedrock LLMs** for dynamic analytics and reporting to a **QuickSight Q chatbot**
- Partnered with senior engineers to **fine-tune** open-source **Transformers (e.g., LLaMA 2 on SageMaker)** using dataset features like **app criticality** and **integration counts**, and defined evaluation metrics (**perplexity, human feedback**) to validate model outputs against **ground truth**
- Extended architecture to incorporate **Azure OpenAI GPT-4** and **GCP Vertex AI PaLM 2** as fallback inference endpoints, orchestrated via **FastAPI microservices** for cost-optimized routing and high availability across cloud providers
- Contributed to sprint planning and peer reviews in **GitHub**, authored **Swagger/OpenAPI specs** for **Gen AI microservices**, and maintained **JIRA stories** to track **prompt engineering** experiments and model iterations
- Built a **computer vision solution** using **PyTorch & Detectron2** on **SageMaker** to automate defect inspection and increase throughput.
- Designed a **predictive maintenance** pipeline ingesting **sensor data** via **Kafka** and **Spark** to forecast equipment failures and trigger proactive alerts
- Developed real-time analytics for **robotic sewing cells**, optimizing cycle timing and tool usage to boost **line efficiency**
- Packaged and deployed models with **MLflow** and **GitHub Actions**, enabling automated **training, testing, and blue-green releases**
- Implemented **GPU-accelerated inference** and **batch tuning strategies** to reduce **latency** and **resource consumption**
- Created interactive **Grafana** and **QuickSight** dashboards to monitor model performance, data drift, and operational KPIs
- Integrated **CI/CD pipelines** across the **ML lifecycle**, from **data ingestion** to **production deployment**
- Collaborated with **stakeholders** to define **data requirements**, establish **success metrics**, and measure **ROI**
- Optimized **cost** and **scalability** by **containerizing services** with **Docker** and orchestrating on **EKS**

➤ **Cognizant Technology Solutions, India (Telangana) – Software Engineer - AI/ML June 2020 – May 2022**

- Developed **ensemble learning models** to **predict patient readmissions** and integrated outcomes into care management workflows
- Built an **NLP chatbot** with **Rasa** to **automate support inquiries**, reducing **manual effort** and improving satisfaction
- Architected secure **AWS pipelines** using **Lambda, Step Functions**, and **S3** to preprocess large **EHR datasets** for **ML training**
- Trained and deployed **CNN classifiers** for **medical image analysis**, accelerating diagnostic triage processes
- Implemented **streaming analytics** to generate **real-time clinical risk alerts** for **sepsis** and other conditions
- Designed and exposed **RESTful APIs** to connect **ML services** with downstream **applications** and **user interfaces**
- Modeled **NoSQL** database schemas in **MongoDB** to support **high-throughput inference workloads**
- Participated in **Agile** ceremonies, producing **UML** documentation, conducting code reviews, and maintaining **sprint backlogs**
- Reduced **cloud costs** by rightsizing **EC2** instances and applying **S3 lifecycle policies** for **data retention management**

- **Cognizant Technology Solutions, India (Telangana) – Software Developer Intern** **Oct 2019 – May 2020**
- Engineered and deployed **RESTful APIs** that seamlessly connected **backend applications** to **AI/ML processing pipelines**, showcasing a firm grasp of modern software architectures
 - Actively participated in **data preprocessing, exploratory analysis, and feature engineering**, directly contributing to the development of **recommendation systems** and **predictive analytics models**
 - Built **NoSQL** database storage solutions tailored for **ML recommendation engines** using **MongoDB**
 - Contributed to **Agile sprints**, provided documentation via **UML** diagrams, and improved **backend maintainability** with dependency injection (**Spring IoC**)
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EDUCATION:

- **University of Texas, Dallas - UTD** May 2024
Master of Science, Business Analytics
- **Gandhi Institute of Technology and Management University, India** May 2020
Bachelor of Science, Computer Science and Engineering

ACADEMIC PROJECTS:

- **EY Data Challenge – Land Classification of Rice Fields:**
- Accessed Sentinel-1 SAR imagery through the Microsoft Planetary Computer API for downstream analysis.
 - Raised model accuracy from 52 % to 86 % by applying targeted feature engineering and ensemble stacking.
 - Applied k-fold cross-validation alongside Bayesian hyper-parameter optimization to bolster model robustness.
 - Delivered fully reproducible Jupyter notebooks and concise demo sessions for stakeholders.